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Memory chip test pad access management to facilitate data security

Abstract

A system for providing memory chip test pad access management to facilitate data security is disclosed. A host issues a command to access a non-volatile memory of a memory chip system via a test pad. A controller acknowledges the command by transmitting a response to the host to authenticate the host for access. The host then issues an authenticated command to modify a reserved byte of a protected memory partition of the non-volatile memory. The controller responds to the authenticated command and the reserved byte is modified. Firmware of the memory chip system monitors the modification of the reserved byte and notifies the memory chip system to activate a switch in an access control unit controlling access to the non-volatile memory. The switch is then activated, thereby closing a circuit to connect the test pad with the non-volatile memory. The host then access the non-volatile memory via the test pad.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a continuation application of Int. Pat. App. No. PCT/CN2022/114753, filed Aug. 25, 2022, the entire disclosure of which application is hereby incorporated herein by reference.

FIELD OF THE TECHNOLOGY
(1) At least some embodiments disclosed herein relate to memory devices in general, and more particularly, but not limited to, a system for providing memory chip test pad access management to facilitate data security.

BACKGROUND

(2) Typically, a computing device or system includes one or more processors and one or more memory devices, such as memory chips or integrated circuits. The memory devices may be utilized to store data that may be accessed, modified, deleted, or replaced. The memory devices may be, for example, non-volatile memory devices that retain data irrespective of whether the memory devices are powered on or off. Such non-volatile memories may include, but are not limited to, read-only memories, solid state drives, and NAND flash memories. Additionally, the memory devices may be volatile memory devices, such as, but not limited to, dynamic or static random-access memories, which retain stored data while powered on, but are susceptible to data loss when powered off. Based on receipt of an input, the one or more processors of the computing device or system may request that a memory device of the computing system retrieve stored data associated with or corresponding to the input. In certain scenarios, the data retrieved from the memory device may include instructions, which may be executed by the one or more processors to perform various operations and may include data that may be utilized as inputs for the various operations. In instances where the one or more processors perform operations based on instructions from the memory device, data resulting from the performance of the operations may be subsequently stored into the memory device for future retrieval.

(3) Test pads, which typically reside on a non-volatile memory chip package, may be utilized to test the operative functionality of the non-volatile memory of the non-volatile memory chip package. For example, test pads may be utilized by a manufacturer of non-volatile memory to test the non-volatile memory to cure defects, modify componentry, and perform modifications to the operative functionality o the non-volatile memory prior to distribution to third parties. While test pads facilitate troubleshooting of memory devices, the use of test pads can result in significant data security issues. Since test pads are typically directly connected to memory pads, hackers can potentially clone user data, tamper with critical system files, or implant backdoor programs to conduct remote system attacks. As a result, functionality associated with accessing non-volatile memories via test pads may be enhanced to provide greater security.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements.

(2) FIG. **1** shows a memory device and host device for supporting memory chip test pad access management in accordance with embodiments of the present disclosure.

(3) FIG. **2** shows a non-volatile memory chip system and host of a system-on-a-chip for supporting memory chip test pad access management featuring an access control unit disconnecting access to non-volatile memory via a test pad in accordance with embodiments of the present disclosure.

(4) FIG. **3** shows a non-volatile memory chip system and host of a system-on-a-chip for supporting memory chip test pad access management featuring an access control unit connecting access to non-volatile memory via a test pad in accordance with embodiments of the present disclosure.

(5) FIG. **4** illustrates test pads connected to non-volatile memory according to embodiments of the present disclosure.

(6) FIG. **5** illustrates a method for facilitating memory chip test pad access management to facilitate data security in accordance with embodiments of the present disclosure.

(7) FIG. **6** illustrates a schematic diagram of a machine in the form of a computer system within which a set of instructions, when executed, may cause the machine to facilitate functionality supporting memory chip test pad access management according to embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) The following disclosure describes various embodiments for memory chip test pad access management to facilitate data security of memory devices. At least some embodiments of the present disclosure relate to non-volatile memories, such as flash memory devices, and securing access to non-volatile memories, such as via test pads. Test pads are devices or components that may be utilized to effectively test and monitor the operative functionality and features of a memory chip. For example, test pads may be components that may reside on a non-volatile memory chip package that may be separated from defined pins of the memory chip (e.g., JEDEC-defined pins). In certain embodiments, tests pads may serve as an accessible channel through which a host (e.g., a processor on the same chip, a processor on a separate chip, a separate device, etc.) may monitor and access non-volatile memory of the memory chip connected to the test pads. When a host has access to the test pads and the test pads are connected to the non-volatile memory, the host may issue commands to write data, read data, modify data, access data, or perform a variety of other operations with respect to the non-volatile memory. Additionally, the host may receive test data resulting from testing of the non-volatile memory conducted by the test pads. The test data may be analyzed and utilized by a host to determine whether the non-volatile memory is operating in a manner as required or expected. In certain embodiments, the host may utilize the test data to facilitate determination as to whether the non-volatile memory is to be provided to a customer, discarded, or modified.

(9) Nevertheless, since test pads typically reside on the non-volatile memory chip package and are directly connected to the non-volatile memory (e.g., NAND pads, eMMC, UFS, memory cards, BGA package SSD, etc.) of the memory chip package, malicious actors may utilize the test pads to clone user data stored on the non-volatile memory. In certain scenarios, such malicious actors may utilize the test pads as a communication channel to tamper or destroy critical system files stored on the non-volatile memory of the memory chip package. In other scenarios, malicious actors may even utilize test pad access to implant backdoor programs into the non-volatile memory to later conduct a remote system attack on a device incorporating the non-volatile memory. Such attacks are often fatal to the device under attack and are not allowed for applications incorporating non-volatile memory. For example, critical components of autonomous vehicles (e.g., advanced drive-assistance systems, dashboards, black boxes, etc.), banking and finance equipment (e.g., point-of-sale terminals, automated teller machines, etc.), and consumer devices (e.g., cellular phones, tablets, gaming devices, etc.) can be commandeered by malicious actors that may have access to test pads having unfettered access to non-volatile memories of memory chips.

(10) According to embodiments of the present disclosure, systems, apparatuses, and methods are disclosed herein that intercept and control a host's access to non-volatile memory (e.g., NAND dice) via test pads of non-volatile memory systems (e.g., chips). In certain embodiments, the systems, apparatuses, and methods may incorporate the use of a unique access control unit that incorporates internal logic that serves as a bridge for signals of non-volatile memories to test pads of non-volatile memory systems and vice versa. For example, the non-volatile memory systems may include an access control unit that incorporates a switch to switch on or off the connection(s) between tests pads and the non-volatile memories connected to the test pads. In certain embodiments, the validity of a command issued by a host may be utilized to switch the connection on or off. In certain embodiments, the systems, apparatuses, and methods may also incorporate the use of secure protocols to facilitate authentication of a host with the non-volatile memories. For example, in certain embodiments, replay protected communication protocols may be utilized with the systems, apparatuses, and methods to provide protection against malicious data tampering that may be attempted by malicious actors. The replay protected memory mechanism may be utilized to prevent illegal attacks through vendor commands issued by host devices and componentry. For example, Replay Protected Memory Blocks may be an exemplary replay protocol that may be utilized to support the functionality of the systems, apparatuses, and methods disclosed herein. In

certain embodiments, a reserved byte in a replay-protected partition of the non-volatile memory may be utilized to serve as an indicator of test pad access control.

(11) Operatively, in certain embodiments, the systems, apparatuses, and methods may support test pad access management in the following exemplary manner. Initially, a host (e.g., system on a chip, processor of the chip, or other host) may issue a dedicated vendor command to a non-volatile memory chip through defined pins (e.g., JEDEC defined pins) according to a replay-protected required communication protocol (e.g., RPMB). For example, in certain embodiments, the host may issue a CMD56 command including an argument. The non-volatile memory system may acknowledge the command (e.g., the CMD56 command) with a response code that may be transmitted back to the host by the non-volatile memory system. In certain embodiments, once the host receives the correct response from the non-volatile memory system, the host may issue an authenticated command (e.g., CMD25, which may be a data write command) with the correct message authentication code (MAC) key and address of a protected memory partition (e.g., RPMB partition) as the argument of the command to change the reserved byte from zero to one (e.g., 0x0 to 0x1). In certain embodiments, the non-volatile memory system may receive the authenticated command and transmit a reply to the host including the correct response to the command. The non-volatile memory system may cause the reserved by type be changed from 0x0 to 0x1, for example.

(12) Firmware of the non-volatile memory system may be configured to monitor the protected memory partition. For example, the firmware may be configured to monitor changes to the reserved byte of the protected memory partition. Upon detection of the value change of the reserved byte, the firmware may be configured to transmit a notification to the non-volatile memory system (e.g., such as to a controller of the non-volatile memory system or directly signal the access control unit) to activate one or more switches in the access control unit. The activation of the switch may cause the switch to close a circuit connecting the test pad(s) to the non-volatile memory. In certain embodiments, the default state of the access control unit is that the test pad(s) are disconnected from the non-volatile memory (e.g., NAND pads) after powering up the non-volatile memory system or device containing the non-volatile memory system. Once the switch(es) of the access control unit is/are closed and the circuit is closed, the host may be configured to access the non-volatile memory via the test pad(s). In certain embodiments, for example, the host may be able to perform various types of signal operations including, but not limited to, issuing commands for writing data to the non-volatile memory, reading data from the non-volatile memory, modifying data in the non-volatile memory, accessing data from the non-volatile memory, or a combination thereof. Additionally, in certain embodiments, the host may be configured to monitor a bus for the non-volatile memory and may be configured to monitor and obtain testing data resulting from testing conducted via the test pads. Such testing data may be analyzed to determine whether the non-volatile memory is operating in an expected manner, whether the non-volatile memory contains any failures, and whether the non-volatile memory need to be repaired, updated, or both.

(13) In certain embodiments, through a similar process flow as described above, the host may also be configured to disconnect the test pads (i.e., cause the switch to open the closed circuit) from the non-volatile memory of the non-volatile memory system after toggling the reserved byte in the protected memory partition back to the original value (e.g., toggling the reserved byte back to 0x0 from 0x1). For example, the host may issue a command with an argument associated with disconnecting access to the non-volatile memory of the non-volatile memory system (e.g., system on a chip, etc.). The non-volatile memory system may acknowledge the command by transmitting a response (e.g., response code) back to the host. Once the host receives the correct response from the non-volatile memory system, the host may be configured to issue an authenticated command (e.g., CMD 25) with the correct MAC key and address of the protected memory partition as the argument to change the reserved byte back to 0x0 from 0x1, for example. The non-volatile memory system may receive the authenticated command and may reply with a correct response back to the host. The non-volatile memory system may the modify the reserved byte and the firmware of the

non-volatile memory system may detect the value change for the reserved byte. The firmware may be configured to notify the non-volatile memory chip to turn off the switches in the access control unit. The non-volatile memory system may then cause the switches of the access control unit to be turned off, thereby causing the circuit connecting the test pad(s) with the non-volatile memory to be opened. Once the circuit is opened, the test pad(s) may be disconnected from the non-volatile memory, thereby preventing the host from accessing the non-volatile memory until the reserved byte is modified back to 0x1, for example. The processes described above may be repeated as necessary to provide or revoke access to the non-volatile memory of the non-volatile memory system via the test pad(s). Based on at least the foregoing, the functionality provided by the embodiments of the present disclosure provide significant enhancement test pad access technologies, while also ensuring increased data security and access control capabilities.
(14) Referring now also to FIG. **1**, FIG. **1** illustrates an exemplary architecture for a memory device **102** and host device **103** that may be utilized to support memory chip test pad access management in accordance with embodiments of the present disclosure. The memory device **102** and other componentry illustrated in the Figures may belong to a system **100**. In certain embodiments, the memory device **102** is, for example, an SSD, eMMC, UFS, memory card, BGA package SSD, or other storage device, or a NAND-based flash memory chip or module that is capable of encoding and decoding stored data, such as by utilizing an encoder **160** and decoder **162** of the memory device **102**. In certain embodiments, the memory device **102** may include any amount of componentry to facilitate the operation of the memory device **102**. In certain embodiments, for example, the memory device **102** may include, but is not limited to including, a non-volatile memory **104**, which may include any number of memory blocks, such as memory block **120** (e.g., Block A), a memory interface **101**, a controller **106** (which may include an encoder **160** and/or a decoder **162**), any other componentry, or a combination thereof. The memory device **102** may communicatively link with a host device **103**, which may be or include a computer, server, processor, autonomous vehicle, any other computing device or system, or a combination thereof.
(15) In certain embodiments, the controller **106** of the memory device **102** may be configured to control access to the non-volatile memory **104**. In certain embodiments, user data **130** is provided by controller **106** to non-volatile memory **104**, such as by utilizing memory interface **101**. For example, the user data may be obtained from the host device **103** to be stored in the non-volatile memory **104**, such as in memory block **120**. In certain embodiments, the controller **106** may include an encoder **160** for generating ECC data (e.g., such as when writing data to the non-volatile memory **104**), and decoder **162** for decoding ECC data (e.g., when reading data, such as from the non-volatile memory **104**).
(16) As indicated above, the memory device **102** may be configured to receive data (e.g., user data) to be stored from host device **103** (e.g., over a serial communications interface, or a wireless communications interface). In certain embodiments, the user data **130** may be video data from a device of a user, sensor data from one or more sensors of an autonomous or other vehicle, text data, audio data, virtual reality data, augmented reality data, information, content, any type of data, or a combination thereof. In certain embodiments, memory device **102** stores the received data in memory cells (not explicitly shown) of non-volatile memory **104**. In one example, the memory cells may be provided by one or more non-volatile memory chips. In one example, the memory chips may be NAND-based flash memory chips, however, any type of memory chips or combination of memory ships may also be utilized. In certain embodiments, the system **100**, memory device **102**, and host device **103** may be integrated or communicatively linked with the system **200** illustrated in FIGS. **2** and **3**, which is discussed in further detail below.
(17) Referring now also to FIGS. **2** and **3**, FIGS. **2** and **3** illustrate a non-volatile memory chip system **210** and host **203** of a chip system **200** (e.g. system-on-a-chip or other chip) for supporting memory chip test pad access management featuring an access control unit for controlling access to non-volatile memory of the non-volatile memory chip system **210** via test pads **240**. In certain

embodiments, the chip system **200** may be incorporated into any type of device including, but not limited to, a computer, a mobile device, a wearable device, an autonomous vehicle, a server, any type of device, or a combination thereof. In certain embodiments, the chip system **200** may include a host device **203**, a non-volatile memory system **210**, and any other componentry that may be included on a chip. In certain embodiments, the host device **203** may be a processor, computing device, or any type of componentry that may serve as a host for the chip system **200**. In certain embodiments, the host device **203** may be attached to the chip system **200** and may be configure to communicate with a non-volatile memory system **210**, which may also reside on the chip system **200**. In certain embodiments, the host device **203** may interact with the non-volatile memory chip system **210**, such as to generate commands to write data to non-volatile memory **220**, **230** contained therein. Other interactions may include, but are not limited to, accessing data from the non-volatile memory **220**, **230**, reading data from the non-volatile memory **220**, **230**, erasing data from the non-volatile memory **220**, **230**, performing any other action with respect to the non-volatile memory **220**, **230**, or a combination thereof. In certain embodiments, the host device **203** may be configured to obtain testing data obtained from test pads **240** conducting tests that test the operative functionality of the non-volatile memory **220**, **230**. In certain embodiments, the host device **203** may issue commands to the non-volatile memory system **210** through defined pins (e.g., JEDEC-defined pins **205**) of the non-volatile memory system **210**. In certain embodiments, the commands that are issued through the defined pins **205** may follow replay-protected required communication protocols (e.g., RPMB protocols).

(18) In certain embodiments, the non-volatile memory system **210** of the chip system **200** may include any number of components. In certain embodiments, the components of the non-volatile memory system **210** may include, but are not limited to, pins **205** (e.g., JEDEC-defined pins **205**), a controller **208**, a non-volatile memory **220**, non-volatile memories **230**, test pads **240**, an access control unit **250**, among other componentry. In certain embodiments, the pins **205** may be utilized by the host **203** to communicate with the non-volatile memory system **210**. In certain embodiments, the controller **208** may be configured to control access to the other componentry of the non-volatile memory system **210**. For example, the controller **208** may control access to the non-volatile memories **220**, **230**, the access control unit **250**, the test pads **240**, other componentry of the non-volatile memory system **210**, or a combination thereof. In certain embodiments, the controller **208** may be configured to receive commands transmitted by or issued by the host device **203** that are intended for conducting operations with respect to the non-volatile memories **220**, **230**, access control unit **250**, or test pads **240**.

(19) The non-volatile memory system **210** may include any number or type of memories including, but not limited to, non-volatile memories **220**, **230**. Illustratively, FIGS. **2** and **3** depict the use of four non-volatile memories: one non-volatile memory **220** (e.g., NAND0), and three non-volatile memories **230** (e.g. NAND1, NAND2, and NAND3). Notably, however, any number of non-volatile memories may be utilized with the non-volatile memory system **210**. In certain embodiments, the non-volatile memories may be any type of memories including, but not limited to, NAND (e.g., NAND pads), SSD, UFS, memory cards, BGA package SSD, and other types of memories. Illustratively, FIGS. **2** and **3** depict the use of NAND memories. In certain embodiments, the non-volatile memory **220** may be configured to include a designated user area portion **222** (e.g., partition) dedicated to storing user data, a designated portion (e.g., partition) for including a replay protected memory block (RPMB) **224**, and a designated portion (e.g., partition) for firmware **226**. The user data stored in the user area **222** may include any type of data that may be stored in the non-volatile memory **220**. Such data may include, but is not limited to, information, content, measurements, instructions, any type of data, or a combination thereof.

(20) In certain embodiments, the replay protected memory block **224** may be configured to protect against replay attacks and may have any desired size. For example, the partition size for the replay protected memory block **224** may be multiples of 128 KB. In certain embodiments, the replay

protected memory block **224** may utilized RPMB protocol to implement authentication and security for the non-volatile memory system **210**, however, in certain embodiments, other security and authentication protocols may be utilized. In certain embodiments, at the manufacturing stage or at other desired times, an authentication key may be written into the replay protected memory block **224** and may be utilized as a shared secret to authenticate interactions and transactions conducted between the non-volatile memory system **210**, the host **203**, and other devices and componentry. In certain embodiments, commands, messages, transactions, or signals may be authenticated with the non-volatile memory system **210** by utilizing Message Authentication Code (MAC). In certain embodiments, MAC may comprise a hash value generated by utilizing the authentication key, a random number (e.g., provided by a host, such as host **203**), and the message itself (or command, signal, etc.) using HMAC SHA-256, for example. In certain embodiments the MAC and key may be utilized to sign all operations, commands, signals, transactions, or messages intended to access the replay protected memory block **224**. In certain embodiments, the replay protected memory block **224** may also include a reserved byte, which may be utilized to dedicatedly serve as an indicator of access control for the test pad **240**. For example, if the reserved byte has a value of 0, this may mean that access to the non-volatile memories **220**, **230** via test pads **240** should not be granted or enabled. If the reserved byte, on the other hand, has a value of 1, this may mean that access to the non-volatile memories **220**, **230** via test pads **240** should be granted or enabled.

(21) In certain embodiments, the firmware **226** may be configured to reside or be installed within the non-volatile memory **220**. In certain embodiments, the firmware **226** may be configured to control the components of the non-volatile memory system **210**. In certain embodiments, the functionality and features provided by the present disclosure may require the full initialization of the non-volatile memory system **210** with the firmware **226** running. In certain embodiments, the firmware **226** may also be configured to monitor the replay protected memory block **224** and the value of the reserved byte stored therein. In certain embodiments, the firmware **226** may reside within its own partition within the non-volatile memory **220**. The firmware **226** may be configured to notify componentry of the non-volatile memory system **210** when a change the reserved byte value occurs. For example, in certain embodiments, the firmware **226** may be configured to notify the controller **208**, the access control unit **250**, or a combination thereof, when the reserved byte changes. Additionally, the firmware **226** may be configured to transmit a signal to the controller **208**, the access control unit **250**, or a combination thereof, to activate the switch **255** to close the circuit connecting the test pads **240** with the nonvolatile memories **220**, **230** so that the host **203** may access the non-volatile memories **220**, **230** via the test pads **240**. Similarly, if the circuit is currently closed and the reserved byte in the replay protected memory block **224** is changed to 0, the firmware **226** may notify the controller **208**, the access control unit **250**, or both to turn off the switch **255** to open the circuit, thereby disconnecting the test pads **240** from the non-volatile memories **220**, **230**. The host **203** may then prevented from accessing the non-volatile memories **220**, **230** via the test pads **240**.

(22) In certain embodiments, the non-volatile memories **230** may be generally configured to include user areas **232**. In certain embodiments, user area **232** may be a partition of the non-volatile memories **230** that may be configured to store user data. In certain embodiments, the host device **230** may be configured to access data, modify data, erase data, or perform other actions with respect to the data by interacting with the non-volatile memories **230** via the test pads **240**. In certain embodiments, the non-volatile memory system **210** may include any number of test pads **240**. The test pads **240** may be configured to test the functionality of the non-volatile memories **220**, **230**, such as during a manufacturing process or prior to distribution to customers. In certain embodiments, the test pads **240** may reside on the non-volatile memory system **210** and may be separated from defined pins (e.g., JEDEC-defined pins), as illustrated in FIG. **4**. The test pads **240** may serve as an accessible channel to facilitate field issue analysis associated with the non-volatile memories **220**, **230** as well. In certain embodiments, the test pads **240** may be configured to

measure test data associated with the operation of the non-volatile memories **220**, **230**, which may then be utilized to determine whether the non-volatile memories **220**, **230** are operating in an expected or desired manner, whether the non-volatile memories **220**, **230** are fixable, whether the non-volatile memories **220**, **230** are corrupted or malfunctioning, or a combination thereof. In certain embodiments, test data generated via the test pads **240** may be provided to the host device **203** for further analysis and processing.
(23) In certain embodiments, the access control unit **250** of the non-volatile memory system **210** may be configured to control access to the switch **255** that may be utilized to open or close the circuit (e.g., by switching ON or OFF the connection) connecting the test pads **240** to the non-volatile memories **220**, **230**. In certain embodiments, the access control unit **250** may be an internal logic unit that serves as a bridge from the signals of the non-volatile memories **220**, **230** (e.g., NAND dice) to the test pads **240** of the non-volatile memory system **210** and vice versa. In certain embodiments, the access control unit **250** may be configured to receive a signal from the firmware **226** or from the controller **208** to activate the switch **225** (i.e., turn ON) to close the circuit, as shown in FIG. **3**. The signal may be received such as after detection by the firmware **226** that the value of the reserved byte of the replay protected memory block **224** has changed to 1 from 0. Similarly, when the reserved byte of the replay protected memory block **224** has changed to 0 from 1, a signal may be received, such as from the firmware **226** or controller **208** to deactivate the switch **225** (i.e., turn OFF) to open the circuit to disconnect the connection between the test pads **240** and the non-volatile memories **220**, **230**. In certain embodiments, the default state of the access control unit **250** upon initialization or power up of the non-volatile memory system **210** may be that the non-volatile memories **220**, **230** are disconnected from the test pads **240** (i.e., the circuit is open), as shown in FIG. **2**. In certain embodiments, the signal to switch on or off the connection between the test pads **240** and non-volatile memories **220**, **230** may be provided only after validating and authenticating a command received from the host device **203**.
(24) Operatively, the system **100**, the system **200**, or a combination thereof, may operate as illustrated by the following exemplary use-case scenario. In the following example, a host (e.g., host device **203**, which may be a system-on-a-chip) may be utilized to access non-volatile memories **220**, **230** of a non-volatile memory system **210** via one or more test pads **240** contained therein. In a default state, the test pads **240** may be disconnected from the non-volatile memories **220**, **230** upon power up, the circuit may be in an opened state, and the reserved byte of the replay protected memory block **224** may be set to 0x0. The host device **203** may issue a command (e.g., CMD56) with an argument, such as 0x110005DB to the non-volatile memory system **210**, such as via defined pins **205**. The non-volatile memory system **210** may acknowledge the command (e.g., CMD56) with a response code (e.g., 0x0D000009003F) back to the host device **203**. After the host device **203** receives the correct response from the non-volatile memory system **210**, the host device **203** may issue or transmit an authenticated command (e.g., a RPMB authenticated data write command (CMD25)) with the correct MAC key and address for the replay protected memory block **224** (e.g., RPMB address) as the argument to change the reserved byte of the replay protected memory block **224** from 0x0 to 0x1.
(25) The non-volatile memory system **210** may receive the authenticated command and may reply with the correct response (e.g., 0x190000090031) to the host device **203**, thereby authenticating and authorizing the host device **203** to communicate with the non-volatile memories **220**, **230** via the test pads **240**. If, however, the key is not correct or the address is incorrect, the host device **203** may be prevented from accessing the non-volatile memories **220**, **230** via the test pads **240**. If the key and address are correct, the reserved byte of the replay protected memory block **224** may be changed or toggled from 0x0 to 0x1. The firmware **226**, which monitors the replay protected memory block **224**, may detect the change in value for the reserved byte and may transmit a notification to the controller **208**, the access control unit **250**, or both to turn on the switch **255** (or switches) to close the circuit. The switch **255** may be turned on and the circuit may then be closed,

thereby connecting the test pads **240** with the non-volatile memories **220**, **230**. Once the circuit is closed, the host device **203** may be configured to perform various signal operations with respect to the non-volatile memories **220**, **230** via the test pads. Such signal operations may include, but are not limited to, issuing commands, monitoring NAND buses, writing data, erasing data, modifying data, accessing data, and the like. Through the same process flow, the host device **203** may be configured to also disconnect the test pads from the non-volatile memories **220**, **230** by causing the reserved byte to be toggled back to 0x0 from 0x1. The processes of activation and deactivation may be repeated as necessary or as desired.

(26) Referring now also to FIG. **5**, FIG. **5** illustrates a method **500** for providing memory chip test pad access management to facilitate data security according to embodiments of the present disclosure. For example, the method of FIG. **5** can be implemented in the system of FIGS. **1-4**, **6** and any of the other systems illustrated in the Figures. In certain embodiments, the method of FIG. **5** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method of FIG. **5** may be performed at least in part by one or more processing devices (e.g., controller **106** of FIG. **1**, controller **208** of FIGS. **2** and **3**, etc.). Although shown in a particular sequence or order, unless otherwise specified, the order of the processes may be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

(27) The method **500** may include steps for enabling and providing memory chip test pad access management for non-volatile memory devices including test pads that are utilized to test the operative functionality of the non-volatile memory. In certain embodiments, the method **500** may be performed by utilizing any combination of the systems **100**, **200**, **600**, by utilizing any combination of the componentry contained therein, or a combination thereof. At step **502**, the method **500** may include receiving a first command from a host (e.g., host **103**, **203**, etc.). In certain embodiments, the first command issued by the host may be a command associated with accessing non-volatile memory (e.g., non-volatile memory **220**, **230**) of a non-volatile memory system (e.g., non-volatile memory chip system **210** of system **200**) via one or more test pads (e.g., test pads **240**). In certain embodiments, the first command may be received by the non-volatile memory system, such as by a controller (e.g., controller **106** or controller **208**) of the non-volatile memory system. For example, the first command may be a CMD56 command (e.g., memory health status command) including an argument.

(28) At step **504**, the method **500** may include transmitting a response acknowledging the first command. For example, in certain embodiments, in response to the CMD56 command, the non-volatile memory system may acknowledge the command with a response code that may be sent to the host. At step **506**, the method **500** may include receiving a second command from the host to modify a reserved byte of a protected memory partition (e.g., RPMB **224**) of the non-volatile memory (e.g., NAND memory pad **220**). In certain embodiments, the second command may be a RPMB authenticated write command (e.g., CMD25) including a message authentication code (MAC) key and RPMB address (e.g., the address of RPMB **224**) as the argument of the command to change the reserved byte from 0x0 to 0x1. In certain embodiments, when the reserved byte is 0, a circuit (e.g., circuit **255**) may be disconnected, thereby disconnecting the connection between the test pads and the non-volatile memory. In certain embodiments, when the reserved byte is set to 1, the circuit may be closed, thereby connecting the test pad(s) with the non-volatile memory.

(29) At step **508**, the method **500** may include determining whether the second command from the host includes the correct key (e.g., MAC key) and address for the protected memory partition. If the correct key or address are not included with the second command, the method **500** may proceed to

step **510**. At step **510**, the method **500** may include preventing the host from accessing the non-volatile memory of the non-volatile memory system via the test pad(s). If however, the second command from the host includes the correct key and the address for accessing the protected memory partition, the method **500** may proceed to step **512**. At step **512**, the method **500** may include transmitting a response to the host in response to the second command. In certain embodiments, the response may be utilized to authenticate the host providing the host's authority to communicate with the non-volatile memory via the test pad(s). For example, the non-volatile memory system may be configured to transmit the correct response to the host. At step **514**, the method **500** may include modifying the reserved byte of the protected memory partition in response to or based on the second command. For example, the modification of the reserved byte may include changing the reserved byte from 0x0 to 0x1.

(30) At step **516**, the method **500** may include receiving a notification from firmware (e.g., firmware **226**) of the non-volatile memory system to activate one or more switches (e.g., switch **255**) of an access control unit (e.g., access control unit **250**) of the non-volatile memory system. In certain embodiments, the firmware may actively monitor the protected memory partition, among other componentry of the non-volatile memory partition, and may be configured to detect any changes in values of the reserved byte of the non-volatile memory partition. Upon detection of the change in value of the reserved byte of the protected memory partition, the firmware may notify the controller of the non-volatile memory system or directly notify the access control unit to activate the switch(es) of the access control unit. At step **518**, the method **500** may include activating the switch(es) in the access control unit to close a circuit connecting the non-volatile memory to the test pads. Once the circuit is closed, the method **500** may proceed to step **520**. At step **520**, the method **500** may include enabling the host to access the non-volatile memory via the test pad. For example, once the circuit is closed and the non-volatile memory is connected to the test pad via the closed circuit, the host may be enabled to perform various types of signal operations with respect to the non-volatile memory and test pads. In certain embodiments, the host may issue commands (e.g., write data to the non-volatile memory, access the non-volatile memory, modify data in the non-volatile memory, etc.), monitor a memory bus (e.g., NAND bus) of the non-volatile memory system, and perform other actions with respect to the non-volatile memory. In certain embodiments, the host may also be utilized to disconnect the test pads from the non-volatile memory by issuing commands to cause the reserved by to toggle from 0x1 to 0x0, which would then cause the switches to be turned off and cause the circuit to be opened. Notably, the method **500** may incorporate any of the other functionality as described herein and may be adapted to support the functionality of the systems **100**, **200**, and **600**.

(31) FIG. **6** illustrates an exemplary machine of a computer system **600** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, can be executed. In certain embodiments, the computer system **600** can correspond to a host system or device (e.g., the host device **203** of FIGS. **2** and **3**) that includes, is coupled to, or utilizes a memory system (e.g., the non-volatile memory system **210** or chip system **200**). In certain embodiments, computer system **600** corresponds to memory device **102**, host device **103**, or a combination thereof. In certain embodiments, the machine may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, and/or the Internet. The machine can operate in the capacity of a server or a client machine in client-server network environment, as a peer machine in a peer-to-peer (or distributed) network environment, or as a server or a client machine in a cloud computing infrastructure or environment. In certain embodiments, the machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, a switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of

instructions to perform any one or more of the methodologies discussed herein.

(32) In certain embodiments, the exemplary computer system **600** may include a processing device **602**, a main memory **604** (e.g., read-only memory (ROM), flash memory, dynamic random-access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), static random-access memory (SRAM), etc.), and/or a data storage system **618**, which are configured to communicate with each other via a bus **630** (which can include multiple buses). In certain embodiments, processing device **602** may represent one or more general-purpose processing devices such as a microprocessor, a central processing unit, or the like. More particularly, the processing device may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets, or processors implementing a combination of instruction sets. In certain embodiments, the processing device **602** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **602** is configured to execute instructions **626** for performing the operations and steps discussed herein. For example, the processing device **602** may be configured to perform steps of the method **500** and support functionality provided by the systems **100** and **200**. In certain embodiments, computer system **600** may further include a network interface device **608** to communicate over a network **620**.

(33) The data storage system **618** can include a machine-readable storage medium **624** (also referred to as a computer-readable medium herein) on which is stored one or more sets of instructions **626** or software embodying any one or more of the methodologies or functions described herein. The instructions **626** can also reside, completely or at least partially, within the main memory **604** and/or within the processing device **602** during execution thereof by the computer system **600**, the main memory **604** and the processing device **602** also constituting machine-readable storage media. The machine-readable storage medium **624**, data storage system **618**, and/or main memory **604** can correspond to the memory device **102**, the non-volatile memory system **210**, or a combination thereof.

(34) Reference in this specification to “one embodiment” “an embodiment” or “certain embodiments” may mean that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. The appearances of the phrases “in one embodiment” and “in certain embodiments” in various places in the specification are not necessarily all referring to the same embodiment(s), nor are separate or alternative embodiments mutually exclusive of other embodiments. Moreover, various features are described which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be requirements for some embodiments, but not other embodiments.

(35) Although some of the drawings illustrate a number of operations in a particular order, operations which are not order dependent may be reordered and other operations may be combined or broken out. While some reordering or other groupings are specifically mentioned, others will be apparent to those of ordinary skill in the art and so do not present an exhaustive list of alternatives. Moreover, it should be recognized that the stages could be implemented in hardware, firmware, software or any combination thereof.

(36) In the foregoing specification, the disclosure has been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A system, comprising: a non-volatile memory configured to store data; an access control unit configured to control access to the non-volatile memory by a host via a test pad; and a controller, wherein the controller is configured to; receive a first command from the host, wherein the first command is associated with accessing the non-volatile memory via the test pad; transmit, to the host, a response code acknowledging the first command; modify, in response to a second command issued based on the response code being determined to be correct, a reserved byte of a protected memory; activate, in response to detection of modification of the reserved byte, a switch in the access control unit to close a circuit to connect the test pad to the non-volatile memory; and enable the host to access the non-volatile memory via the test pad after activation of the switch.
2. The system of claim 1, wherein the system further comprises firmware and the controller is further configured to activate the switch in the access control unit in response to a notification received from the firmware of the non-volatile memory.
3. The system of claim 2, wherein the firmware is configured to: monitor the reserved byte; and generate the notification upon detecting that the reserved byte of the protected memory partition of the non-volatile memory has been modified.
4. The system of claim 1, wherein the controller is further configured to: receive the second command to modify the reserved byte of the protected memory partition of the non-volatile memory, wherein the second command to modify the reserved byte of the protected memory partition comprises a replay protected memory block authenticated data write command, wherein the replay protected memory block authenticated data write command comprise a message authentication code key and a replay protected memory block address associated with the protected memory partition of the non-volatile memory.
5. The system of claim 1, wherein the controller is further configured to modify the reserved byte by changing a zero value of the reserved byte to a one value.
6. The system of claim 1, wherein the controller is further configured to receive a third command from the host, wherein the third command is associated with disconnecting access to the non-volatile memory via the test pad.
7. The system of claim 6, wherein the controller is further configured to transmit, to the host, a response acknowledging the third command.
8. The system of claim 7, wherein the controller is further configured to receive a fourth command from the host to modify the reserved byte of the protected memory partition.
9. The system of claim 8, wherein the controller is further configured to modify the reserved byte of the protected memory partition based on the fourth command.
10. The system of claim 9, wherein the controller is further configured to deactivate a switch in the access control unit to open a circuit to disconnect the test pad from the non-volatile memory.
11. The system of claim 10, wherein the controller is further configured to prevent the host from accessing the non-volatile memory via the test pad after deactivation of the switch.
12. The system of claim 1, wherein the controller is further configured to receive a signal from the host to monitor an operation conducted by the non-volatile memory via the test pad.
13. The system of claim 1, wherein the controller is configured to facilitate full initialization of the non-volatile memory and a firmware associated with the non-volatile memory.
14. A method, comprising: receiving, by a controller of a non-volatile memory system, a first command from a host, wherein the first command is associated with accessing non-volatile memory of the non-volatile memory system via a test pad; transmitting, by the controller of the non-volatile memory system and to the host, a response code acknowledging the first command; modifying, by the controller of the non-volatile memory system and in response to a second command issued based on the response code being determined to be correct, a reserved byte of a

protected memory partition; activating, by the controller of the non-volatile memory system and in response to detection of modification of the reserved byte, a switch in an access control unit to close a circuit to connect the test pad to the non-volatile memory; and enabling, by the controller of the non-volatile memory system, the host to access the non-volatile memory via the test pad after activation of the switch.

15. The method of claim 14, further comprising authenticating the host with the non-volatile memory system based on the response acknowledging the first command.

16. The method of claim 14, further comprising enabling the host to write data to the non-volatile memory via the test pad, read data from the non-volatile memory via the test pad, change data in the non-volatile memory via the test pad, or a combination thereof.

17. The method of claim 14, further comprising enabling the host to obtain a test result associated with the test pad testing the non-volatile memory.

18. The method of claim 14, further comprising activating the switch in response to receiving a notification from a firmware of the non-volatile memory system.

19. The method of claim 14, further comprising opening the circuit in response to a third command received from the host.

20. A device, comprising: a host; a non-volatile memory configured to store data; an access control unit configured to control access to the non-volatile memory by the host via a test pad; and a controller, wherein the controller is configured to; receive a command from the host, wherein the command is associated with accessing the non-volatile memory via the test pad; transmit, to the host, a response code acknowledging the command, wherein the response code facilitates authentication of the host to communicate with the non-volatile memory; modify, in response to a second command issued based on the response code being determined to be correct, a reserved byte of a protected memory partition; activate, in response to detection of modification of the reserved byte, a switch in the access control unit to close a circuit to connect the test pad to the non-volatile memory; and enable the host to access the non-volatile memory via the test pad upon closing the circuit.